



Linear Systems Theory

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Module 3

Lecture 1

Math Preliminaries: Linear Algebra 5



Types of Matrices

Types of Matrices

- ▶ Square Matrices : size $n \times n$
- ▶ Rectangular matrix : size $m \times n$
- ▶ Special cases of Square matrices:
 - Identity matrix : I_n
 - Symmetric matrices : $A = A^T$
 - Skew-Symmetric matrices : $A = -A^T$
 - Diagonal matrices : $a_{ij} = 0 \forall i \neq j$
 - Lower Triangular matrix : $a_{ij} = 0 \forall i < j$
 - Upper Triangular matrix : $a_{ij} = 0 \forall i > j$
 - Orthogonal matrix : $AA^T = I$



Change of Basis

Change of basis refers to a transformation which transforms coordinates of a vector w.r.t. one basis to coordinates w.r.t. another basis

- ▶ The set of basis vectors for a vector space are not unique
- ▶ A vector space can be represented using many different sets of basis vectors
- ▶ Different basis vectors have different advantages and hence it is useful to be able to shift from one basis to another which can be represented as a linear transformation
- ▶ The transformation matrix depends on the bases of the vector spaces and is referred to as **change of basis matrix** or **change of coordinates matrix**.



Change of Basis/Coordinates Matrix

- ▶ Let B and C be 2 sets of ordered basis for an n -dimensional vector space V

$$B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$$

$$C = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$$

- ▶ Let $\mathbf{x}$ be a vector in V with coordinate representations $\mathbf{x}^B$ and $\mathbf{x}^C$ w.r.t. B and C respectively. What is the transformation or matrix that gives $\mathbf{x}^C$ given $\mathbf{x}^B$?
- ▶ $\mathbf{x}$ is the representation w.r.t. standard basis. We can write $\mathbf{x}$ in terms of basis vectors B and C i.e.,

$$\mathbf{x} = x_1^B \mathbf{b}_1 + \dots + x_n^B \mathbf{b}_n$$

$$\mathbf{x} = x_1^C \mathbf{c}_1 + \dots + x_n^C \mathbf{c}_n$$

- ▶ Equate (1) and (2) and put them in matrix form



Change of Basis/Coordinates Matrix

$$\begin{bmatrix} \vdots & \vdots & & \vdots \\ \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_n \\ \vdots & \vdots & & \vdots \end{bmatrix} \begin{bmatrix} x_1^B \\ x_2^B \\ \vdots \\ x_n^B \end{bmatrix} = \begin{bmatrix} \vdots & \vdots & & \vdots \\ \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_n \\ \vdots & \vdots & & \vdots \end{bmatrix} \begin{bmatrix} x_1^C \\ x_2^C \\ \vdots \\ x_n^C \end{bmatrix}$$

$$\mathbf{B}\mathbf{x}^B = \mathbf{C}\mathbf{x}^C$$

$$\implies \mathbf{x}^C = \mathbf{C}^{-1}\mathbf{B}\mathbf{x}^B$$

- ▶ Therefore, the change of coordinates matrix is $\mathbf{P} = \mathbf{C}^{-1}\mathbf{B}$
- ▶ $\mathbf{P}$ represents a linear transformation $f: V \rightarrow V$ which depends on the sets of basis vectors B and C
- ▶ If $\mathbf{B}$ is a standard basis then, $\mathbf{P} = \mathbf{C}^{-1} \implies \mathbf{x}^C = \mathbf{C}^{-1}\mathbf{x}^B = \mathbf{C}^{-1}\mathbf{x}$
- ▶ If $\mathbf{C}$ is a standard basis then, $\mathbf{P} = \mathbf{B} \implies \mathbf{x}^C = \mathbf{B}\mathbf{x}^B = \mathbf{x}$



Change of Basis - Example

- Eg. Given sets of basis $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $C = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \end{bmatrix} \right\}$ for $\mathbb{R}^2$. What is $\mathbf{x}^C$ given $\mathbf{x}^B = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$?

Exercise 1

Given sets of basis $B = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \end{bmatrix} \right\}$ and $C = \left\{ \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$ for $\mathbb{R}^2$. What is $\mathbf{x}^C$ given $\mathbf{x}^B = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$?



Similarity Transformation

Similar Matrices and Similarity Transformation

Two $n \times n$ matrices **A** and **B** are said to be similar if there exists a non-singular matrix **P** such that:

$$\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

where the matrix **P** is called modal matrix

- ▶ Similar matrices arise because a linear transformation $f: V \rightarrow V$ can be represented by different matrices for different choice of basis
- ▶ Similar matrices represent the same linear transformation $f: V \rightarrow V$ with different sets of basis for V
- ▶ Eg: $f: V \rightarrow V$ is a similarity transformation represented by 2 similar matrices **A** and **B** with respect to two sets of basis for V



Properties of Similar Matrices

- ▶ A matrix is similar to itself with $\mathbf{P} = \mathbf{I}$
- ▶ **Commutative:** If $\mathbf{A}$ is similar to $\mathbf{B}$, then $\mathbf{B}$ is similar to $\mathbf{A}$
- ▶ **Associative:** If $\mathbf{A}$ is similar to $\mathbf{B}$ and $\mathbf{B}$ is similar to $\mathbf{C}$, then $\mathbf{C}$ is similar to $\mathbf{A}$
- ▶ If $\mathbf{A}$ is similar to $\mathbf{B}$, then they have
 1. Equal rank
 2. Equal determinant
 3. Equal trace
- ▶ The above properties can be used as necessary conditions for similarity of matrices



Summary: Mod 3 Lecture 1

- ▶ Change of Basis
- ▶ Similarity Transformation

Contents: Mod 3 Lecture 2

- ▶ Invariant Subspaces
- ▶ Eigen Values and Eigen Vectors

